Dual-Domain Optimization Model Based on Discrete Fourier Transform and Frequency-Domain Fusion for Remote Sensing Single-Image Super-Resolution

Lei Shi, Yubin Cheng, Runrui Li, Hexi Wang, Jun Zhao, Yan Qiang, and Juanjuan Zhao

***Abstract*— Deep neural network models generally enhance super-resolution (SR) reconstruction of remote sensing images but may distort feature edge details. Recovering low-resolution (LR) remote sensing images with clear texture and high-fidelity edge details is challenging. Recent approaches improve feature fusion in the frequency domain, yielding promising results. We propose a dual-domain optimization network (DDOM) based on discrete Fourier transform (DFT) and frequency-domain complex-valued neural networks. Unlike end-to-end approaches in the image domain, DDOM incorporates frequency-domain information via DFT transformation operators, preserving high-level semantics (phase) and low-level statistical information (magnitude). Using lightweight complex neural networks and Swin Transformer architecture, the frequency-domain and image-domain subnetworks are designed. The dual-domain data consistency constraints ensure the positivity of model optimization. Extensive experiments show superior performance over existing methods in quantitative and qualitative evaluations. The proposed scheme's robustness is verified on additional datasets. Code and model configurations are available at http://github.com/YB-Cheng/DDOM.**

***Index Terms*— Consistency constraints, deep complex networks, discrete Fourier transform (DFT), dual domain, remote sensing, super-resolution (SR), transformer.**

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I. Introduction

GEOGRAPHY information can support human activities in areas such as natural resources, agriculture, and industry. Remote sensing image data are common geographic information [1]. Remote sensing imaging applications place high demands on the high quality and resolution of image data. Remote sensing images are acquired by the sensors mounted on remote sensing platforms. Differences in sensor types and imaging principles determine that imaging schemes basically include photographic imaging, scanning imaging, and radar imaging [2]. Remote sensing applications face significant challenges in spatial resolution due to natural complexities and sensor limitations. To address data issues like poor quality and low resolution (LR), enhancing remote sensing images with super-resolution (SR) techniques has gained substantial attention. Many approaches have been proposed to improve SR reconstruction of remote sensing images, aiming to overcome inherent imaging sensor defects.

Previously, to mitigate the adverse effects of insufficient resolution in remote sensing images on human activities, traditional interpolation-based algorithms were introduced to alleviate blurring resulting from image magnification. These methods primarily emphasize pixel smoothing in remote sensing images, including nearest neighbor, bilinear, and bicubic interpolation [3], [4], [5], [6]. Traditional methods compute the original pixel details of the high-resolution (HR) image corresponding to the LR image by designing a basis function or interpolation kernel. Interpolation-based methods have significant advantages in terms of image processing speed and algorithm complexity [7]. The shortcomings of this type of approach are also more obvious. When the pixel details of the image change greatly, the processing scheme cannot maintain the original edge and texture characteristics of the image and cannot accurately restore the original high-frequency information of the image [8]. Image reconstruction is another method used for LR image recovery, and remote sensing SR reconstruction is a common application area. Tao et al. [10] proposed a research work on the reconstruction of remote sensing images based on stochastic resonance, where the high- and low-frequency information of the image is first filtered out by the discrete wavelet transform and then combined with an interpolation method to improve the wavelet coefficient

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image. Considering SR image reconstruction as a typical low-level vision problem, Hong and Zhang [11] proposed an image reconstruction method based on hidden Markov and wavelet transforms for improving degraded image quality. Jin-liang et al. [12] proposed to design weight factors for wavelet coefficient maps of an image to improve image recovery and reconstruction. The common conventional SR reconstruction algorithms have limited performance in the problem of optimizing the reconstruction error. Meanwhile, good reconstruction results have been obtained by adding constraints and introducing a priori information, such as convex set projection method, maximum a posteriori probability method, Bayesian analysis method, and iterative inverse projection schemes [13], [14], [15], [16]. However, the convergence of the models incorporating prior knowledge is not satisfactory.

When the performance of traditional interpolation algorithms and a priori-based reconstruction algorithms reached a bottleneck on remote sensing SR reconstruction work, artificial intelligence technology made a big splash in various research fields, and AI-driven remote sensing image analysis entered a new era. At the same time, the research on remote sensing image SR reconstruction based on deep learning began to advance rapidly. Due to the fact that convolutional operations enhance the capabilities of the underlying computer vision, Dong et al. [17] first proposed the use of convolutional neural networks (CNNs) for image SR reconstruction work. Compared to the traditional scheme, the SRCNN model was proposed for image end-to-end task with superior performance. Liebal and Körner [18] first proposed the use of SRCNN for remote sensing image SR reconstruction task and recreated a remote sensing dataset based on Sentinel-2 satellite data. Since then, there have been a lot of breakthroughs in the research of remote sensing image SR reconstruction based on deep learning. Kim et al. [19] designed a deep network VDSR trained with LR image and HR image data pairs. Qin et al. [20] and Xu et al. [21] successively proposed CNN with multiscale feature extraction and hopping connectivity for the SR task of remote sensing images. It not only reduces the loss of computational resources but also improves the multiscale feature fusion. After He et al. [22] proposed residual networks, Pan et al. [23] designed residual networks with dense jump connections in combination with an inverse projection network to improve the deep feature reuse for hypersegmented reconstruction of remotely sensed images. For the study of residual learning, Ma et al. [24] proposed to combine wavelet transform to introduce image frequency-domain features, expanding the perspective of deep learning-based modeling. However, its complex network structure and excessive data-driven nature also lead to its high data dependency. Deep learning models designed based on the attention mechanism perform better on computer vision tasks, which can help the field of remote sensing image reconstruction to recover the surface details more finely [25]. Recently, unsupervised learning methods have attracted attention due to their minimal demand for training data, achieving SR by utilizing the information inherent in the images themselves. In particular, networks based on spectral unmixing, by reconceptualizing the HSI-SR task as estimating the abundances and endmembers of unknown HR HSI, have shown potential. However, these methods have led to unnecessary spectral–spatial distortions under the unsupervised framework due to inadequate network structure design and lack of proper constraints. To address these issues, Li et al. [50] and Gao et al. [51] have proposed various improvement strategies, including the design of novel network architectures and loss functions, to enhance the performance and generalization capabilities of HSI-SR. These advancements have provided new perspectives and technical approaches for the field of remote sensing image processing [52], [53], [54], [55]. The proposed and generalized attention mechanism makes deep learning models perform better on feature refinement and high-order feature generation tasks [26], [27], [28]. Feature fusion across frequency and image domains is essentially the same as feature mining guided by the attention mechanism. Both are designed to enhance the deep model's understanding of remote sensing data features. However, the lack of attention to data consistency maintenance while introducing image frequency-domain features leads to the destruction of the realism of image reconstruction results in terms of pixel details. This is the main problem to be solved in the current research on SR reconstruction of remote sensing images incorporating frequency-domain features. The work on exploiting the potential of frequency-domain features of remote sensing images is still not deeply studied. Ledig et al. [29] proposed SRGAN, which is the first time that the generative adversarial network (GAN) model is used for SR reconstruction of remote sensing images. With the help of perceptual loss and training strategies, GAN has achieved excellent results in domain-specific tasks and has driven the generation of more research work on GAN, such as GAN based on the Transformer design [30] and M-GAN with multiple attention guidance [31]. Multisensor, multitemporal, and multifactor remote sensing imagery characteristics have driven research advances regarding weakly supervised learning and unsupervised learning [32], [33]. Among the most recent contributions to the field, Chaouai and Tamaazousti [48] introduced CertSR, which enhances the robustness of real-world SR through the median randomized smoothing technique. However, it exhibits a limited upper performance ceiling in cross-application tasks, such as remote sensing SR. Korkmaz et al. [49] developed WGSR, which employs a loss function in the wavelet domain for model training, effectively managing artifacts generated during SR. Nonetheless, this approach overlooks the cross-domain consistency constraint. The challenges posed by the inadequate performance of existing methods in specific tasks and the preservation of cross-domain data consistency are increasingly evident. To address these issues, we propose a network model for frequency-domain-assisted image resolution restoration, utilizing a discrete Fourier transform (DFT) operator specifically designed for remote sensing image SR reconstruction. Our model demonstrates superior efficiency and performance compared to existing approaches while effectively improving the identified problems. Our contributions are summarized as follows.

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1) We design a deep network based on complex domains, which acts as a pre-stage network and reduces the number of core components of the residual Swin Transformer network (RSTN). The resource loss problem and computational efficiency problem caused by stacking a large number of core components are avoided.
2) We expand the feature extraction perspective of remote sensing image SR reconstruction by fusing feature information such as phase and magnitude in the frequency domain for the image domain based on the DFT operator. This will not require any additional training to perform target domain alignment.
3) We introduce consistency constraints across image and frequency domains to maintain the positivity of the optimization process. Our experiments show that this approach guides the iterative process from the image to the frequency domain, reducing pixel detail distortion in end-to-end tasks. This is the first generalization of the application of dual-domain data consistency constraints in the SR reconstruction of remote sensing images.

II. Methods

In this section, the main theory of our proposed modeling scheme is explained and illustrated. The structure of the proposed dual-domain remote sensing image SR reconstruction model with DFT and frequency-domain assisted optimization is shown in Fig. 1. The structure diagram contains the data flow and the structural details of the main modules. The model consists of four components, which are the data transformation operator based on DFT (inverse DFT), the depth complex subnetwork in the frequency domain, and the Swin Transformer subnetwork in the image domain. After the work of the frequency-domain subnetwork module is completed, the image-domain subnetwork is trained for the optimization of cross-domain information interaction under the optimization objective of dual-domain data consistency constraints. Further explanation of the modules in the proposed model is described as follows.

A. Fourier Frequency-Domain Transformation Operator

For image information expressed digitally, real values often represent only pixel differences. Depth models based on computer vision are unable to tap into the special semantics present in the frequency domain of an image. Pixel information can be represented as a superposition of a series of frequency components through DFT. These frequency components can be used to characterize different spatial features in an image such as edges, textures, and periodic structures. When dealing with complex image signals or data, DFT has shown promising results in image processing across data domains [34]. Learning domain-specific knowledge of the frequency domain of digital images holds promise for improving performance in low-level image recovery tasks. Inspired by Xu et al. [35], we introduce a new data enhancement domain for remote sensing image SR reconstruction. Fourier phase information contains high-level semantics and is less susceptible to domain bias. In order to force the model to capture the phase information, we design a transformation operator based on DFT theory to convert the digitally expressed image information into frequency-domain information. This is a novel perspective for SR reconstruction of remotely sensed images.

Unlike single-channel grayscale images, digital representations of remotely sensed images are often in the form of three channels (RGB). Suppose that a remote sensing image of size 256×256 is digitally expressed as a matrix $I_{256\times256\times3}$. DFT of a color image is usually converted to a grayscale map first, but this process will make the original data lose some pixel information, so the color image is considered as a superposition of three single-channel grayscale maps here. The value of a pixel point at position (x, y) in one of the channels is $f(x, y)$. The 2D-DFT process for one of the color channels of I is

$$\mathcal{F}(u, v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)e^{-2\pi j(ux+vy)}dxdy \tag{1}$$

$$I^{(f)} = \mathrm{dft}(I) \tag{2}$$

where $\mathcal{F}(u, v)$ is the complex value in the frequency domain under that channel. The datametric expression $I^{(f)}$ of the image in the frequency domain is obtained by performing a 2D-DFT of the three channels of the RGB image. After DFT is performed, the data in the frequency domain are usually arranged in such a way that the zero frequency is in the upper left corner. In order to understand the frequency-domain information more intuitively and for further processing, the proposed transformation operator also contains a shift operation

$$I^{(f)} = \mathrm{Shift}\left(I^{(f)}\right) = \mathcal{F}(u - 128, v - 128). \tag{3}$$

It serves to move the zero frequency component to the center of the spectrum. Its corresponding reverse process

$$I^{(\mathcal{F}*)} = \mathrm{Shift}^{-}\left(I^{(f*)}\right) = \mathcal{F}(u + 128, v + 128) \tag{4}$$

$$\mathcal{F}(u, v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)e^{2\pi j(ux+vy)}dudv \tag{5}$$

$$I = \mathrm{idft}(I^{(f)}). \tag{6}$$

The process of discrete Fourier inverse conversion is subject to only a minuscule rounding error, which is negligible in practical calculations. In the following, T and T^{-} will be used to represent the DFT operator and the inverse transformation operator (iDFT operator). The derivability of the operators is an important prerequisite for our next step of constructing dual-domain networks and cross-domain consistency constraints, which stems from the Riemann integrable condition for the DFT function [9].

B. Complex Deep Network

Complex values possess a series of good properties in expressing vector aggregation, rotation, and exponential operations that real values do not have [36], which is one of the motivations for us to design operators based on DFT. Most neural networks are designed and implemented based on real values, and the general strategy to mine complex-valued features through deep network models is to simply separate the real part from the imaginary part or to consider

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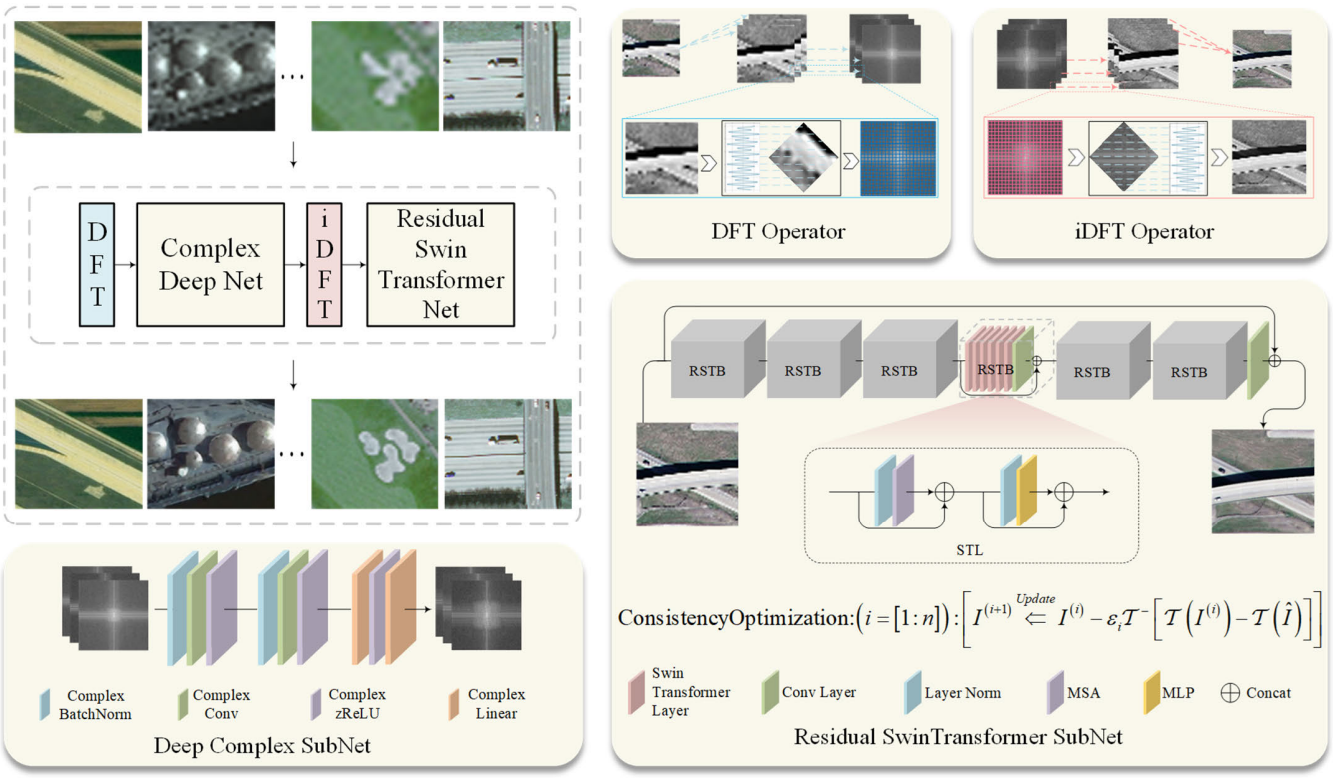


Fig. 1. Proposed model structure diagram and module structure diagram. The model structure diagram is located in the dashed box. Module structure diagrams are located in the other boxes.

only the magnitude and phase angle, but both of them are highly prone to lose the original feature relationships of the complex values. The frequency-domain subnetwork in the proposed model is a complex neural network constructed to leverage complex-valued computation principles, incorporating operations such as convolution, activation, and batch normalization on complex input data. This initial subnetwork enhances remote sensing data recovery and streamlines subsequent recovery stages. By reducing parameters and resource consumption, this approach improves computational efficiency and allows the model to reduce the number of Swin Transformer layers (STLs) in later stages, thus mitigating efficiency issues associated with excessive layers.

Suppose that a complex value is $z = a + bi$, and the real and imaginary parts are different in a logical sense. Set up a complex convolution kernel

$$\omega = A + iB. \tag{7}$$

Then, the convolution process for complex values is as follows:

$$\omega * z = (A * a - B * b) + i(B * a + A * b). \tag{8}$$

In a practical realization, the numerical relations expressed in a matrix form are then available

$$\begin{bmatrix} (\omega * z)^{(r)} \\ (\omega * z)^{(i)} \end{bmatrix} = \begin{pmatrix} A & -B \\ B & A \end{pmatrix} * \begin{pmatrix} a \\ b \end{pmatrix}. \tag{9}$$

Under Cauchy–Riemann equations, the minimizability of the network's complex parameters ensures that the model's objective and activation functions are required for backpropagation in the complex convolutional layer [36]. An activation function shaped like the real activation operation [37] takes the form

$$\text{zReLU}(\omega * z) = \begin{cases} \omega * z, & \text{if } \vartheta_z \epsilon \left[0, \dfrac{\pi}{2}\right] \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

$$\begin{cases} (\omega * z)^{(r)} > 0, & (\omega * z)^{(i)} = 0 \\ (\omega * z)^{(i)} > 0, & (\omega * z)^{(r)} = 0 \end{cases} \tag{11}$$

The complex function differentiability under this strategy restricts only to cases of the form (111). In practical operations, cases like (11) do not occur. Therefore, the zReLU function is chosen as the activation function of the model.

Unlike batch normalization in the real value domain, batch normalization for complex values requires normalizing the complex values to a normal complex distribution. The data can be scaled by taking the square root of the variances along the two principal components of complex-valued data separately. The batch normalization scheme for complex values is shaped like

$$\tilde{z} = \frac{z - E(z)}{\sqrt{C}} \tag{12}$$

$$C = \begin{bmatrix} C^{(rr)} & C^{(ri)} \\ C^{(ir)} & C^{(ii)} \end{bmatrix} = \begin{bmatrix} \text{Conv}(z^{(r)}, z^{(r)}) & \text{Conv}(z^{(r)}, z^{(i)}) \\ \text{Conv}(z^{(i)}, z^{(r)}) & \text{Conv}(z^{(i)}, z^{(i)}) \end{bmatrix} \tag{13}$$

where C is a 2×2 covariance matrix. Similar to real-valued batch normalization, learnable shift parameter β and scale parameter γ are also used in complex-valued batch normalization. Complex-valued batch normalization

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is defined as

$$\text{ComplexBN}(\tilde{z}) = \gamma\tilde{z} + \beta. \tag{14}$$

ComplexBN establishes a specific link for the real and imaginary parts of $\tilde{z}$, which will avoid to a greater extent the overfitting of the two parts in co-optimization [38].

C. Residual Swin Transformer

Swin Transformer is a deep learning model based on the Transformer architecture [39]. The efficiency of the model is improved by the design of the shift window mechanism and local attention mechanism. The image-domain subnetwork proposed in this article constitutes a residual Swin Transformer block (RSTB) by adding additional convolutional layers and residual connections at the end of each Swin Transformer block. A convolutional layer and residual connections are also included after the multiple RSTB shown. The design of the additional convolutional layer will help to enhance the translational equivariance of Swin Transformer. The design of the residual connection will improve the multilevel feature fusion capability of the original model [40].

As shown in Fig. 1, the image is fed into the image-domain subnetwork by first passing through a shallow feature extraction layer, which is used to extract the shallow features of the under-resolved image. Given an under-resolved image I_{LR}

$$f^{(0)} = \text{ConvL}(I_{\text{LR}}). \tag{15}$$

I_{LR} will be fed into a convolutional layer. The convolutional kernel size is 3×3. It is used to get the input for the first RSTB. This layer will be used for shallow feature extraction.

The main structure of the image-domain subnetwork consists of multiple RSTBs (the number is determined by the actual experimental setup). The design of the RSTB consists of six STLs and a convolution layer. As shown in Fig. 1, each STL contains the multihead self-attention mechanism design (MSA), the multilayer perceptron (MLP), and the layer normalization (LN).

Under the design of the MSA mechanism, given an input $f^{(i)}$ with dimensions $H \times W \times C$, the model reshapes the input into a feature of $(HW/M^2) \times M^2 \times C$. M^2 is the number of windows. The self-attention calculation scheme for each window is as follows:

$$\begin{cases} Q = f_{\text{local}}^{(i)} P_Q \\ K = f_{\text{local}}^{(i)} P_K \\ V = f_{\text{local}}^{(i)} P_V \end{cases} \tag{16}$$

$$\text{SA}(Q, K, V) = \text{SoftMax}\left(\frac{QK^{\text{T}}}{\sqrt{d}} + B\right)V \tag{17}$$

where $f_{\text{local}}^{(i)}$ is one of the local window features; K, Q, and V are query, key, and value matrices, respectively; and P_Q, P_K, and P_V are the shared projection matrices between different windows. $\text{SA}(Q, K, V)$ is the self-attention matrix of the local window features. B is the relative position encoding. Execute h attention in parallel and finally concatenate the multiple attention results. In addition, with the design of the multilayer perceptron, the features can be further transformed. The multilayer perceptron contains two fully connected layers and an intermediate GELU layer. Both structures contain an LN operation before designing and residual jump connections are added to them. The whole process can be viewed as

$$f^{(i)'} = \text{MSA}\left(\text{LN}\left(f^{(i)}\right)\right) + f^{(i)} \tag{18}$$

$$f^{(i)''} = \text{MLP}\left(\text{LN}\left(f^{(i)'}\right)\right) + f^{(i)'}. \tag{19}$$

D. Dual-Domain Data Consistency Constraints

The direction of iterative learning of network models follows the loss minimization principle. The optimization process is prone to corrupting local pixel values within the constraints of the cumulative error minimization objective. This may lead to oversmoothing of local texture details of the image or giving up the fidelity of local pixel features to maintain the overall similarity of the image. The inspiration for the dual-domain consistency constraint comes from K-Space [42]. K-Space is the bearing space for image data with frequency space as the coordinate system. It actually belongs to the frequency space, which can also be called Fourier space. Deep learning-based network models, during pretraining, often employ common pretraining objectives as constraints. Constructing dual-domain consistency constraints based on the spatial and frequency domains can help enhance the performance ceiling of the model during pretraining. This also aids in strengthening the model's feature learning capabilities. The primary task of this module is to concentrate the objectives of dual-domain optimization through specific fusion strategies. This will contribute to enhancing the model's backpropagation process.

The traditional gradient descent algorithm is shaped as

$$I^{(i+1)} \overset{\text{Update}}{\Longleftarrow} I^{(i)} - \varepsilon_i (I^{(i)} - \hat{I}) \tag{20}$$

where $I^{(i)}$ is the generated result during the ith round of image-domain subnetwork training, $\hat{I}$ is the ground truth, $I^{(i+1)}$ is the $(i+1)$th round of training result, and ε_i is the learning optimization scale factor for the ith round of training. Due to the extreme sensitivity of the image pixel space to spatial changes in the frequency domain, we design a dual-domain data consistency constraint algorithm for remote sensing images

$$\text{Train}(\text{epoch}i = [1:n]) \quad \left[I^{(i+1)} \overset{\text{Update}}{\Longleftarrow} I^{(i)} - \varepsilon_i \mathcal{T}^{-}[\mathcal{T}\left(I^{(i)}\right) - \mathcal{T}(\hat{I})]\right] \tag{21}$$

where $\mathcal{T}/\mathcal{T}^{-}$ is the DFT operator and the inverse operator. The loss is calculated by converting the generated results and ground truth of the current training round to the frequency domain through the operator. The inverse operator returns the frequency-domain loss to the image domain to be used as a guide for image training optimization. Optimizing the training of the image-domain model by the frequency-domain loss will help to enhance the fidelity of the image details. The model optimization objectives under the dual-domain data consistency constraint algorithm are as follows:

$$\text{MIN}\mathcal{L} = \left\|\mathcal{T}^{-}[\mathcal{T}\left(I^{(i)}\right) - \mathcal{T}(\hat{I})]\right\|_2^2. \tag{22}$$

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TABLE I
WORKFLOW OF DDOM

Algorithm 1: Proposed DDOM

Pre-Train

Initialization:
  ·Initialize $\mathcal{T}/\mathcal{T}^{-}$ parameters
  ·Initialize CDN parameters
  ·Initialize RSTN parameters
Pre-Train CDN with $\mathcal{T}(I)$
While Eq (22), do:
  IF epoch < n:
    Pre-Train RSTN with $\mathcal{T}^{-}(RSTN(\mathcal{T}(I)))$ as Eq (21)
    save&update save

Validate

DDOM.load
DDOM.eval
$I^{(\oint')} = CDN[\mathcal{T}(I^{(Valid)})]$
$\hat{I} = RSTN[\mathcal{T}^{-}(I^{(\oint)})]$

The differentiability of the DFT process ensures the feasibility of backpropagation of this loss in real training.

The workflow of the proposed model is summarized in Table I (Algorithm 1).

III. EXPERIMENT

In this section, first, the source of the dataset and the data processing strategy used to validate the performance of our model are explained. In addition, the details of the experimental process are explained. Finally, the results of all the experiments are analyzed and presented.

A. Data and Training

Two datasets were used for the performance validation work of the proposed model. The two datasets are UC Merced Remote Sence Dataset [43] and NWPU-RESISC45 Remote Sence Dataset [44]. UC Merced Remote Sence Dataset is a dataset of remotely sensed imagery on documented land use for scientific research. The dataset was released by UC Merced CV Lab in 2010. The dataset contains 21 categories of ground scenes with 100 images of each ground scene, totaling 2100 images for the entire dataset. The images were manually extracted by the USGS from the National Map's urban area image collection. The image size in the dataset is 256 × 256, and the image pixel resolution is 1 ft. The NWPU-RESISC45 Remote Sence Dataset is a large-scale remote sensing imagery dataset published by NWPU in 2016. The dataset contains 45 classes of ground scenes with 700 images of each ground scene, totaling 31 500 images for the entire dataset. The size of the dataset is 256 × 256. We downsampled the image data in the two datasets using bicubic interpolation, and then, the size recovery was used to construct the simulated low-resolution (LR) dataset (the downsampling factors are ×2, ×4, and ×8). In addition, in the size recovery, in order to simulate the impact of objective factors such as simulated image acquisition equipment and natural environment on remote sensing imaging, we also set the Gaussian noise calculation parameter mean and variance to be 0.01 and 0.05, respectively, to simulate the generation of Gaussian noise [45]. In the experimental work, the NWPU-RESISC45 dataset is used for the training of the proposed model. The UC Merced dataset is not involved in the model training and is used for the robustness validation of the proposed model. Since the number of ground scene images in each category in the dataset used for training is consistent, it is ensured that the number of images involved in training, validation, and testing is consistent for all categories. For each category, 560 instances of images were randomly selected, totaling 25 200 image data pairs for model pretraining. Among the remaining images, 70 examples of each category image were randomly selected, totaling 3150 image data pairs for model validation. In the remaining images of each category, 70 examples, totaling 3150 image data pairs were used for model testing.

The two complex-valued convolutional layers of the deep complex subnetwork have a convolutional kernel size of 5 × 5, a step size of 2, and a number of output channels of 20 and 50. The 500 hidden neurons are set up in the fully connected layer to construct the specification size mapping. A random initialization scheme is used to out pool the weights in the convolutional layer. Training was performed by the SGD optimization algorithm with the parameter momentum set to 0.99. The learning rate was decreased from the initial $5e^{-4}$ to $1e^{-4}$ in two steps. The network completed convergence when the number of training rounds reached 70. The task faced by residual Swin Transformer is a light image task, so work [40] uses four RSTBs, each containing six STLs and one convolutional layer. The number of attention heads is set to 6. The window size is 8, the number of channels is 60, and the size of the convolutional kernel is 3 × 3. The Adam optimization algorithm is used for training, where $\beta_1 = 0.9$ and $\beta_2 = 0.99$. The learning rate decreases progressively from $1e^{-4}$ to $2e^{-5}$. The batchsize of the dual-domain subnetwork is set to 48 and 4. All experiments are performed on a PC in Python 3.8 + PyTorch1.7.1 environment (PC configuration is Ubuntu 20.04 + Intel Xeon Silver 4210 R CPU + 64G RAM + two NVIDIA RTX A5000 GPUs). Due to the large number of parameters and computation, we also used PyTorch's parallel computing strategy in order to speed up the completion of training.

B. Comparison Experiments

To verify the performance of the proposed model (DDOM), this section tests the proposed model against other schemes, including Bicubic [6], SRCNN [18], VDSR [19], DSSR [46], PCRCAN [28], CertSR [48], WGSR [49]. Bicubic is the most commonly used image resolution restoration scheme among the traditional interpolation schemes. SRCNN first realizes the deep learning based on building a three-layer CNN for the image SR end-to-end task. VDSR designs deep CNN with one jump connection to enhance deep feature information perception by using CNN as a backbone. DSSR introduces an attention mechanism into CNN and improves the model performance by learning low-frequency features of the image. CNN is introduced to assist image reconstruction by constructing

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TABLE II

QUANTITATIVE EVALUATION (MEAN ± STANDARD DEVIATION) OF THE PROPOSED MODEL AND DIFFERENT SCENARIOS USING THE NWPU-RESISC45 DATASET AT SCALE = 2. BOLD VALUES INDICATE THE BEST RESULTS. "AVERAGE OF ALL" DENOTES THE OVERALL AVERAGE ACROSS ALL CATEGORIES, WHILE OTHER ELEMENTS REPRESENT THE AVERAGE VALUE OF A SPECIFIC CATEGORY

Model / Scene Class	Bicubic	SRCNN	VDSR	PCRBPN	DSSR	CertSR	WGSR	Purposed	Scale ×2
Average of All	27.4178±0.5582	30.4640±0.3020	30.9434±0.8994	31.6272±0.2882	32.1449±0.8714	32.7218±0.9866	32.9912±0.5615	**33.5356±0.1322**	**PSNR**
	0.8292±0.0030	0.8648±0.0096	0.8640±0.0016	0.9012±0.0079	0.9197±0.0048	0.9279±0.0098	0.9248±0.0042	**0.9338±0.0101**	**SSIM**
airplane	25.5478±0.6214	28.2127±0.5443	28.5441±0.4540	29.9705±0.3558	30.4422±0.2168	31.3426±0.6464	31.6330±0.8054	**31.9322±0.1510**	**PSNR**
	0.8036±0.0040	0.8594±0.0047	0.8571±0.0034	0.8814±0.0056	0.8958±0.0047	0.9054±0.0055	0.9137±0.0042	**0.9194±0.0113**	**SSIM**
beach	23.3657±0.5206	27.6628±0.8051	27.5829±0.9632	28.7549±0.6115	29.7061±0.9539	30.1158±0.1483	30.3218±0.5566	**30.5771±0.2329**	**PSNR**
	0.7835±0.0017	0.8241±0.0094	0.8413±0.0095	0.8884±0.0036	0.8976±0.0076	0.9131±0.0010	0.9127±0.0021	**0.9102±0.0096**	**SSIM**
chaparral	26.8221±0.9899	30.3788±0.5716	30.7574±0.9267	31.1179±0.6910	32.5123±0.3016	33.6193±0.1194	33.3694±0.8346	**34.1524±0.3133**	**PSNR**
	0.8078±0.0038	0.8678±0.0061	0.8852±0.0052	0.9042±0.0021	0.9114±0.0096	0.9238±0.0045	0.9283±0.0055	**0.9366±0.0053**	**SSIM**
commercial area	23.1899±0.5644	26.5920±0.5602	26.2511±0.4334	27.1956±0.5443	27.4977±0.2888	27.9754±0.5794	28.1328±0.4429	**28.6134±1.0103**	**PSNR**
	0.7872±0.0094	0.8417±0.0022	0.8671±0.0042	0.8890±0.0095	0.8723±0.0041	0.8774±0.0048	0.8847±0.0035	**0.8941±0.0032**	**SSIM**
freeway	24.3911±0.3062	27.2266±0.1486	27.6259±0.6166	28.2626±0.2160	29.3214±0.7967	29.7136±0.7103	30.0459±0.8092	**30.3122±0.9588**	**PSNR**
	0.7978±0.0096	0.8434±0.0081	0.8590±0.0016	0.8798±0.0042	0.8878±0.0054	0.8950±0.0034	0.8954±0.0086	**0.9011±0.0115**	**SSIM**
island	25.2091±0.6041	28.2976±0.7505	28.5646±0.4621	30.8225±0.9023	31.5326±0.3530	32.1645±0.9537	33.2735±0.9642	**33.6715±1.2635**	**PSNR**
	0.8084±0.0031	0.8510±0.0095	0.8851±0.0051	0.9032±0.0057	0.9192±0.0073	0.9253±0.0091	0.9311±0.0076	**0.9312±0.0094**	**SSIM**
overpass	24.8229±0.7108	28.5521±0.6619	28.3499±0.9077	29.6902±0.9189	30.5796±0.4381	30.5950±0.1093	30.6743±0.1570	**31.1613±1.3314**	**PSNR**
	0.7949±0.0032	0.8557±0.0051	0.8610±0.0048	0.8828±0.0064	0.8987±0.0023	0.9052±0.0019	0.9094±0.0042	**0.9109±0.0104**	**SSIM**
runway	28.8864±0.6973	32.1761±0.2543	32.8647±0.6896	34.3313±0.8119	35.1279±0.5213	35.7677±0.7714	36.7813±0.1219	**36.9497±1.6281**	**PSNR**
	0.8327±0.0033	0.9070±0.0064	0.9073±0.0096	0.9207±0.0041	0.9237±0.0097	0.9317±0.0093	0.9383±0.0050	**0.9403±0.0095**	**SSIM**
storage tank	23.9459±0.5280	27.8528±0.8325	27.9059±0.2638	29.9177±0.1774	30.9853±0.2383	31.1714±0.4845	31.6255±0.6210	**32.3394±0.8159**	**PSNR**
	0.7877±0.0087	0.8519±0.0038	0.8757±0.0053	0.8981±0.0023	0.9051±0.0043	0.9148±0.0094	0.9170±0.0016	**0.9267±0.0034**	**SSIM**
wetland	22.4178±0.5582	25.9640±0.3020	25.8434±0.8994	27.2272±0.2882	28.1449±0.8714	29.2218±0.9866	29.2749±0.3926	**29.6492±1.3615**	**PSNR**
	0.7762±0.0030	0.8248±0.0096	0.8564±0.0016	0.8712±0.0079	0.8877±0.0048	0.8979±0.0098	0.8967±0.0067	**0.9015±0.0087**	**SSIM**

TABLE III

QUANTITATIVE EVALUATION OF THE PROPOSED MODEL AND DIFFERENT SCENARIOS USING THE NWPU-RESISC45 DATASET AT SCALE = 4

Model / Scene Class	Bicubic	SRCNN	VDSR	PCRBPN	DSSR	CertSR	WGSR	Purposed	Scale ×4
Average of All	22.7536±0.8653	25.2274±0.1768	25.4804±0.5649	27.3593±0.9289	27.4759±0.3710	27.4763±0.4842	28.1908±0.3597	**28.5133±0.3861**	**PSNR**
	0.7024±0.0060	0.7474±0.0068	0.7460±0.0055	0.7637±0.0050	0.7689±0.0067	0.7784±0.0081	0.7836±0.0097	**0.7862±0.0061**	**SSIM**
airplane	19.3334±0.4283	22.5329±0.9144	22.9936±0.9583	24.2434±0.6074	25.9572±0.5359	26.7467±0.9741	27.1957±0.8732	**27.1963±0.4515**	**PSNR**
	0.6921±0.0028	0.7338±0.0071	0.7382±0.0053	0.7549±0.0095	0.7594±0.0047	0.7583±0.0055	**0.7775±0.0083**	0.7746±0.0061	**SSIM**
beach	20.2612±0.2967	23.2122±0.3738	23.5254±0.7098	25.7666±0.4584	25.5645±0.7081	26.8346±0.9220	26.1755±0.3709	**26.4891±0.6167**	**PSNR**
	0.6977±0.0016	0.7320±0.0013	0.7363±0.0035	0.7522±0.0055	0.7584±0.0047	0.7559±0.0033	**0.7782±0.0025**	0.7716±0.0051	**SSIM**
chaparral	22.9971±0.4273	25.5275±0.8144	26.9020±0.7931	28.4502±0.9465	28.8647±0.9219	29.0383±0.7620	29.4132±0.4450	**29.6749±0.9415**	**PSNR**
	0.7013±0.0043	0.7515±0.0062	0.7582±0.0070	0.7783±0.0039	0.7710±0.0053	0.7749±0.0034	0.7856±0.0068	**0.7912±0.0049**	**SSIM**
commercial area	19.7339±0.7695	22.1496±0.8012	22.6133±0.2676	23.5827±0.6337	23.6256±0.6484	23.6555±0.5267	23.3877±0.1523	**23.9814±0.8434**	**PSNR**
	0.6769±0.0013	0.7110±0.0070	0.7111±0.0050	0.7340±0.0063	0.7399±0.0065	0.7323±0.0065	0.7505±0.0087	**0.7513±0.0045**	**SSIM**
freeway	19.7750±0.1428	22.1966±0.6611	22.6292±0.3425	23.9441±0.1922	23.6200±0.2555	25.8035±0.2661	25.3278±0.9869	**26.6417±0.5411**	**PSNR**
	0.6953±0.0040	0.7449±0.0020	0.7467±0.0061	0.7677±0.0086	0.7642±0.0078	0.7643±0.0031	0.7742±0.0023	**0.7795±0.0075**	**SSIM**
island	22.2919±0.2509	25.9874±0.1522	26.1518±0.5272	27.4252±0.1149	27.9318±0.2152	28.2007±0.4135	**28.9792±0.4126**	28.9641±0.8416	**PSNR**
	0.6999±0.0093	0.7452±0.0038	0.7436±0.0085	0.7679±0.0019	0.7640±0.0062	0.7617±0.0013	0.7749±0.0081	**0.7802±0.0050**	**SSIM**
overpass	20.4468±0.4733	23.8144±0.9191	24.1560±0.1873	25.1248±0.1818	25.6962±0.8798	26.3552±0.3503	**26.9050±0.5243**	26.8616±0.6591	**PSNR**
	0.6963±0.0060	0.7486±0.0098	0.7486±0.0062	0.7657±0.0067	0.7666±0.0027	0.7621±0.0030	0.7753±0.0010	**0.7754±0.0066**	**SSIM**
runway	24.6729±0.3371	27.3498±0.6987	27.6295±0.4764	28.9261±0.3627	28.7760±0.9001	29.1056±0.7042	29.8839±0.5232	**30.1322±0.4417**	**PSNR**
	0.7058±0.0015	0.7576±0.0097	0.7678±0.0037	0.7851±0.0052	0.7855±0.0090	0.7871±0.0073	0.8098±0.0061	**0.8131±0.0104**	**SSIM**
storage tank	24.1923±0.5669	27.1471±0.4062	27.3141±0.8555	28.5125±0.5564	28.2472±0.8579	28.5235±0.5139	29.3363±0.5098	**29.9416±0.9154**	**PSNR**
	0.7047±0.0028	0.7556±0.0047	0.7655±0.0020	0.7878±0.0099	0.7858±0.0081	0.7880±0.0062	0.7979±0.0070	**0.7991±0.0020**	**SSIM**
wetland	18.7536±0.8653	21.9274±0.1768	22.4804±0.5649	23.9440±0.6864	24.3759±0.3710	24.4763±0.4842	24.1895±0.3585	**24.4895±0.6189**	**PSNR**
	0.6924±0.0060	0.7374±0.0068	0.7360±0.0055	0.7497±0.0013	0.7589±0.0067	0.7584±0.0081	0.7562±0.0063	**0.7605±0.0031**	**SSIM**

dense residual blocks and wide feature attention blocks to fuse high-dimensional information features. PCRCAN uses RCAB as the basic block of the model in order to reduce the difficulty of the model and realizes the end-to-end task of image reconstruction in a parallel structure. CertSR improves the robustness of real-world SR through the median randomized smoothing technique, and its performance on task domains is worth noticing. WGSR is worth noticing as an optimization scheme across data domains by using a loss function in the wavelet domain to train the model.

1) Quantitative Analysis: The quantitative PSNR and SSIM calculations for the reconstruction of the proposed model (DDOM) with other schemes at different scale factor settings are shown in Tables II–IV. Compared to the other schemes, the data results of the experiments show that the proposed model achieves the optimal reconstruction results on the NWPU-RESISC45 dataset. This indicates that the proposed model shows better scenario applicability in specific tasks. Compared with WGSR, which has the best reconstruction results except for the proposed model, the proposed model achieves an average PSNR increment of 0.5444, 0.3225, and 0.2431 dB, and an average SSIM increment of 0.0090, 0.0026, and 0.0048 for the three experimental settings. It is interesting to note that the quantitative enhancement of the two evaluation metrics is not significant when the scale factor is 8. This is due to the fact that images with a large degree of undersampling have lost a lot of structural and textural information in the corresponding simulated experimental task scenarios.

2) Qualitative Analysis: Examples of the effects of the proposed model (DDOM) and other schemes for reconstruction with different scale factor settings are shown in Figs. 2–4. Bicubic, as a classical scheme for image resolution restoration,

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TABLE IV

QUANTITATIVE EVALUATION OF THE PROPOSED MODEL AND DIFFERENT SCENARIOS USING THE NWPU-RESISC45 DATASET AT SCALE = 8

Model / Scene Class	Bicubic	SRCNN	VDSR	PCRBPN	DSSR	CertSR	WGSR	Purposed	Scale ×8
Average of All	18.8573±0.2939	22.4501±0.9277	22.7437±0.3689	23.5308±0.4235	23.4607±0.4205	23.7370±0.3028	23.8894±0.1069	**24.1325±0.6197**	PSNR
	0.4718±0.0036	0.5371±0.0037	0.5383±0.0082	0.5421±0.0028	0.5489±0.0087	0.5427±0.0031	0.5473±0.0062	**0.5521±0.0034**	SSIM
airplane	17.3025±0.5164	21.3153±0.6701	21.8728±0.2854	21.9074±0.2671	21.9276±0.9345	21.7929±0.5613	21.8271±0.5045	**22.4644±0.4662**	PSNR
	0.4683±0.0047	0.5241±0.0064	0.5228±0.0050	0.5382±0.0013	0.5379±0.0022	0.5328±0.0057	0.5366±0.0075	**0.5382±0.0047**	SSIM
beach	17.2375±0.9473	21.1516±0.2552	21.2630±0.9012	22.5191±0.2459	22.2171±0.9561	22.5469±0.7246	23.4632±0.2309	**23.8246±0.8574**	PSNR
	0.4696±0.0083	0.5279±0.0073	0.5252±0.0048	0.5329±0.0028	0.5346±0.0092	0.5317±0.0052	0.5348±0.0057	**0.5403±0.0073**	SSIM
chaparral	19.2587±0.2467	23.3890±0.9688	23.5341±0.2351	24.5435±0.8311	24.5650±0.9112	24.8347±0.6481	25.4114±0.9731	**25.4269±0.6403**	PSNR
	0.4827±0.0075	0.5510±0.0012	0.5594±0.0050	0.5638±0.0026	0.5641±0.0017	0.5616±0.0045	0.5652±0.0056	**0.5666±0.0035**	SSIM
commercial area	15.8091±0.7363	18.4045±0.8565	18.4961±0.9346	18.9862±0.4622	18.2109±0.7381	18.3399±0.8911	18.9976±0.1019	**19.3341±0.4499**	PSNR
	0.4422±0.0025	0.4949±0.0070	0.4909±0.0011	0.4920±0.0020	0.4986±0.0085	0.4990±0.0050	0.4972±0.0036	**0.4986±0.0060**	SSIM
freeway	18.2470±0.1050	21.5869±0.2008	22.8987±0.1203	23.2832±0.5719	23.3501±0.5330	23.2961±0.3312	**24.5926±0.4437**	24.3124±0.1218	PSNR
	0.4715±0.0059	0.5275±0.0069	0.5369±0.0028	0.5461±0.0096	0.5459±0.0020	0.5450±0.0079	0.5510±0.0086	**0.5517±0.0065**	SSIM
island	23.7636±0.7620	26.9352±0.7499	26.5882±0.4433	27.4493±0.9771	27.2579±0.8034	27.5921±0.4685	**28.8423±0.3398**	28.8069±0.6130	PSNR
	0.507±0.0023	0.5258±0.0068	0.5297±0.0014	0.5392±0.0089	0.5322±0.0063	0.5327±0.0012	0.5454±0.0031	**0.5456±0.0046**	SSIM
overpass	18.8715±0.7676	21.3941±0.4582	22.1246±0.6109	23.6103±0.1609	23.4612±0.1617	23.6345±0.8508	**24.7518±0.2240**	24.3893±0.9595	PSNR
	0.4732±0.0056	0.5251±0.0020	0.5391±0.0089	0.5461±0.0065	0.5442±0.0058	0.5477±0.0082	0.5551±0.0032	**0.5580±0.0045**	SSIM
runway	23.4809±0.4479	25.5723±0.3098	25.7648±0.3032	26.8609±0.7962	26.1815±0.4377	26.3148±0.3461	27.5103±0.2896	**27.1629±0.1960**	PSNR
	0.5052±0.0067	0.5489±0.0094	0.5563±0.0011	0.5610±0.0068	0.5615±0.0089	0.5616±0.0019	0.5731±0.0043	**0.5740±0.0073**	SSIM
storage tank	22.5864±0.4132	24.7880±0.6202	24.8520±0.2760	25.2897±0.2841	25.6566±0.8477	25.3499±0.8013	26.3413±0.4777	**26.6249±0.9608**	PSNR
	0.5068±0.0067	0.5416±0.0023	0.5479±0.0032	0.5568±0.0025	0.5516±0.0074	0.5564±0.0040	0.5654±0.0082	**0.5690±0.0035**	SSIM
wetland	18.4750±0.3087	20.9128±0.5935	21.8434±0.7635	22.9773±0.5150	22.7607±0.4205	22.7370±0.3028	23.9637±0.6144	**23.3712±0.4275**	PSNR
	0.4738±0.0077	0.5092±0.0096	0.5175±0.0011	0.5212±0.0029	0.5289±0.0087	0.5227±0.0031	0.5319±0.0081	**0.5350±0.0063**	SSIM

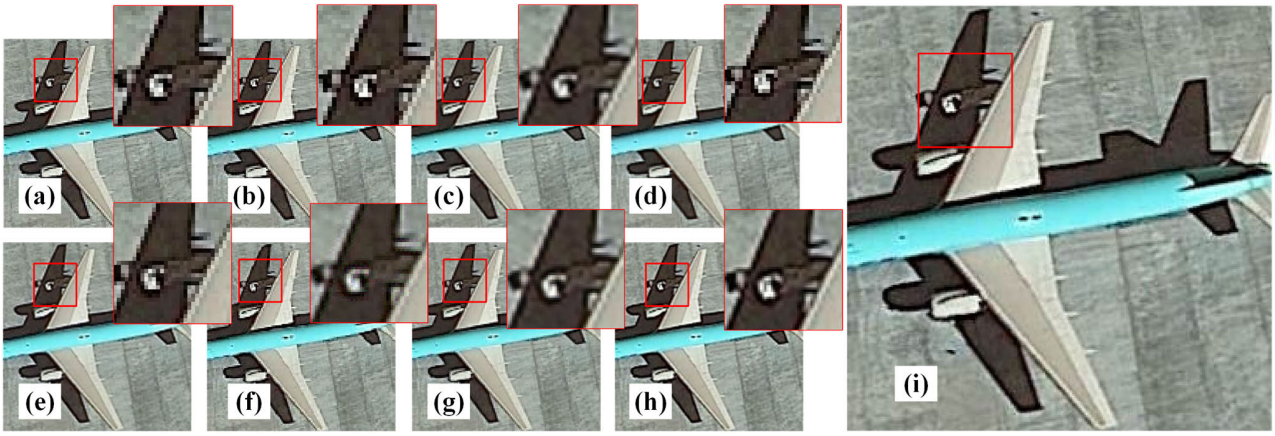


Fig. 2. Comparison experiments on the NWPU-RESISC45 dataset at Scale = 2. The figure is shown as airplane. (a) Bicubic, (b) SRCNN, (c) VDSR, (d) DSSR, (e) PCRBPN, (f) CertSR, (g) WGSR, (h) proposed model, and (i) reference image.

is less effective in reconstructing the edges and textures of the feature targets when the undersampling scale is large. SRCNN and VDSR are smoother for the reconstruction of feature targets under the corresponding constraint targets. WGSR enhances the edge reconstruction effect of the model by introducing frequency-domain features through wavelet transform, and PCRBPN and DSSR both show better ability in local reconstruction, but the data distortion in detail reconstruction should not be ignored. In visual effect, only a very small amount of detail loss occurs. The proposed model significantly improves the detail distortion in the reconstruction process through the two-stage network model and the cross-domain consistency constraints for the image domain and performs well in the image detail fidelity.

C. Robustness Experiments

In order to verify the robustness of the proposed model (DDOM) and the comparison scheme, in this section, the proposed model and the comparison scheme are, respectively, subjected to SR reconstruction effect validation experiments on the UC Merced Remote Sence Dataset, which is not involved in model training.

1) Quantitative Analysis: The quantitative PSNR and SSIM calculations for robustness validation of the proposed model against other schemes with different scale factor settings are shown in Tables V–VII. Compared to the other schemes, the data results of the experiments show that the proposed model achieves the optimal robustness validation results on the UC Merced dataset. This indicates that the proposed model exhibits better robustness in specific tasks. The model-driven Bicubic scheme did not produce significant performance degradation due to the change in the target task scenario. The remaining schemes all produced performance degradation to some extent. Compared to WGSR, which has the best reconstruction results except for the proposed model, the proposed model has an average PSNR increment of 0.3662, 0.2642, and 0.1354 dB, and an average SSIM increment of 0.0044, 0.00013, and 0.0063 for the three experimental setups.

2) Qualitative Analysis: The reconstruction results of the proposed model (DDOM) and other schemes in the robustness

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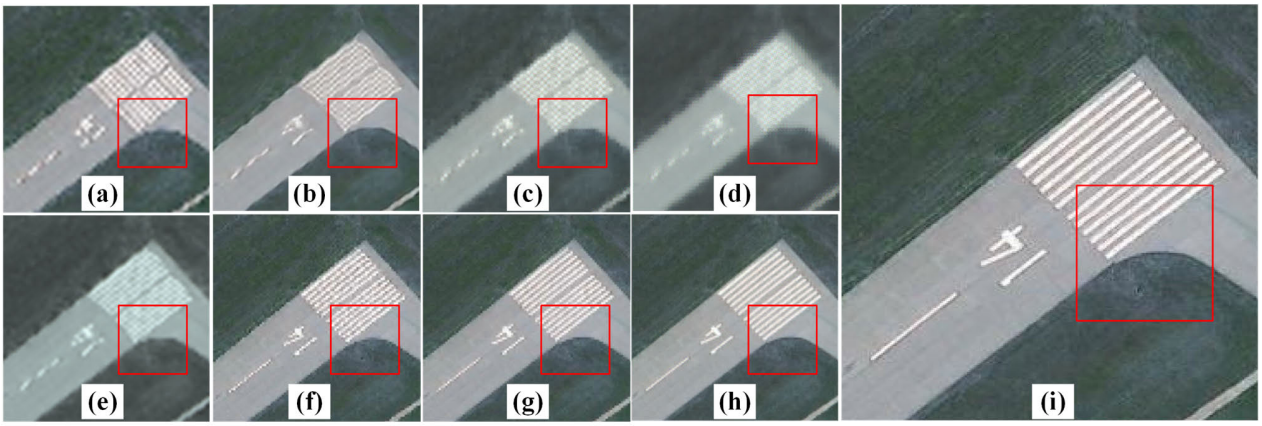


Fig. 3. Comparison experiments on the NWPU-RESISC45 dataset at Scale = 4. The figure is shown as runway. (a) Bicubic, (b) SRCNN, (c) VDSR, (d) DSSR, (e) PCRBPN, (f) CertSR, (g) WGSR, (h) proposed model, and (i) reference image.

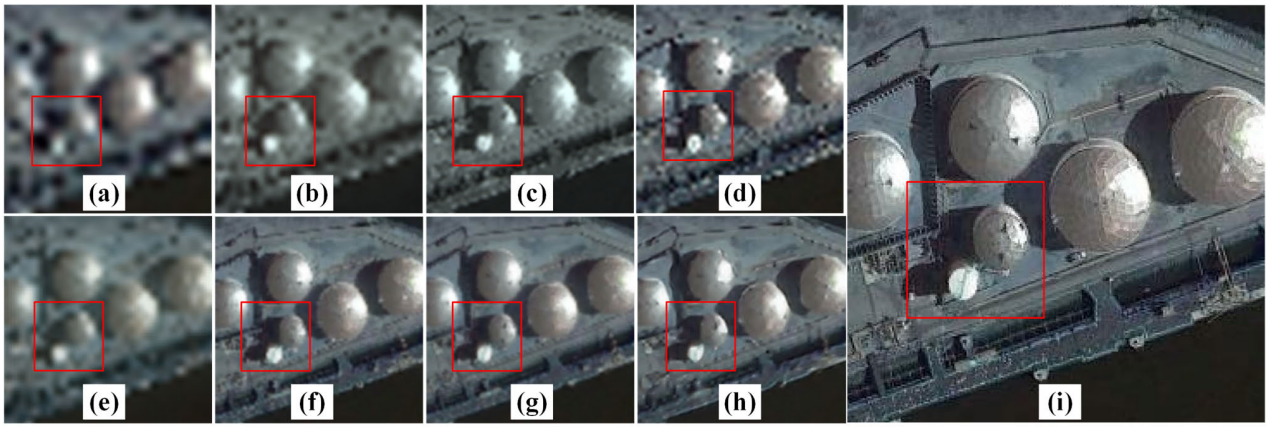


Fig. 4. Comparison experiments on the NWPU-RESISC45 dataset at Scale = 8. The figure is shown as storage tank. (a) Bicubic, (b) SRCNN, (c) VDSR, (d) DSSR, (e) PCRBPN, (f) CertSR, (g) WGSR, (h) proposed model, and (i) reference image.

TABLE V

ROBUSTNESS QUANTITATIVE EVALUATION (MEAN ± STANDARD DEVIATION) OF THE PROPOSED MODEL AND DIFFERENT SCENARIOS USING THE UC MERCED DATASET AT SCALE = 2. BOLD VALUES INDICATE THE BEST RESULTS. "AVERAGE OF ALL" DENOTES THE OVERALL AVERAGE ACROSS ALL CATEGORIES, WHILE OTHER ELEMENTS REPRESENT THE AVERAGE VALUE OF A SPECIFIC CATEGORY

Model / Scene Class	Bicubic	SRCNN	VDSR	PCRBPN	DSSR	CertSR	WGSR	Purposed	Scale ×2
Average of All	27.4268±0.3682	28.3736±0.7985	28.3342±0.3786	30.1087±0.3524	30.1459±0.7078	30.7750±0.8176	30.8018±0.5849	**31.1680±0.7518**	**PSNR**
	0.8359±0.0064	0.8771±0.0087	0.8784±0.0034	0.8935±0.0079	0.8965±0.0097	0.9075±0.0045	0.9083±0.0078	**0.9127±0.0019**	**SSIM**
agricultural	27.7211±0.4016	28.5985±0.7647	28.1107±0.9038	30.1799±0.1610	30.9238±0.6446	30.9705±0.1273	30.2089±0.9921	**31.4009±0.8634**	**PSNR**
	0.8367±0.0094	0.8796±0.0022	0.8712±0.0041	0.8993±0.0046	0.9061±0.0018	0.9065±0.0067	0.9094±0.0085	**0.9148±0.0026**	**SSIM**
baseballdiamond	27.7849±0.4592	28.1057±0.1854	28.2946±0.4757	30.7187±0.9881	30.2252±0.2756	30.5860±0.8002	30.1783±0.5216	**31.1042±0.1668**	**PSNR**
	0.8391±0.0046	0.8754±0.0088	0.8764±0.0063	0.9050±0.0069	0.8996±0.0072	0.9070±0.0053	0.9086±0.0039	**0.9120±0.0047**	**SSIM**
denseresidential	23.1127±0.7776	24.1923±0.9424	24.8958±0.6485	26.5834±0.5744	26.7745±0.9516	27.7792±0.7914	27.8210±0.1925	**27.9383±0.5589**	**PSNR**
	0.6529±0.0075	0.7912±0.0044	0.79±0.0070	0.8611±0.0041	0.8643±0.0060	0.8711±0.0089	0.8786±0.0042	**0.8868±0.0095**	**SSIM**
golfcourse	30.6610±0.2667	32.6852±0.7974	32.6118±0.5534	34.5683±0.4909	34.2131±0.4912	35.0779±0.5379	35.1574±0.9430	**35.2802±0.4901**	**PSNR**
	0.8562±0.0049	0.9219±0.0088	0.9290±0.0031	0.9374±0.0032	0.9365±0.0026	0.9381±0.0028	0.9368±0.0097	**0.9450±0.0033**	**SSIM**
mediumresidential	24.9187±0.5525	25.4795±0.1332	25.4833±0.6990	27.6052±0.5041	27.1183±0.8386	27.1745±0.4322	27.3674±0.4866	**27.6350±0.3166**	**PSNR**
	0.6718±0.0054	0.8192±0.0031	0.81±0.0037	0.8755±0.0016	0.8727±0.0061	0.8800±0.0033	0.8811±0.0082	**0.8818±0.0058**	**SSIM**
overpass	28.8309±0.7046	29.3606±0.4680	29.9488±0.5510	31.1316±0.3756	31.9865±0.6706	32.1042±0.5273	32.0644±0.3499	**32.1641±0.6282**	**PSNR**
	0.8391±0.0020	0.8813±0.0032	0.8875±0.0024	0.9132±0.0096	0.9153±0.0018	0.9110±0.0013	0.9177±0.0044	**0.9218±0.0093**	**SSIM**
runway	28.9796±0.3745	29.4979±0.9599	29.6152±0.8414	31.1838±0.9622	31.8551±0.7215	33.1518±0.3659	33.4015±0.7314	**33.4844±0.7651**	**PSNR**
	0.8456±0.0055	0.8828±0.0014	0.8836±0.0044	0.9016±0.0040	0.9058±0.0041	0.9195±0.0094	0.9277±0.0023	**0.9282±0.0047**	**SSIM**
storagetanks	28.4192±0.5463	30.1291±0.1322	30.6816±0.2849	32.2834±0.8445	32.6332±0.8722	32.7342±0.1978	33.3334±0.7489	**33.5793±0.2698**	**PSNR**
	0.8431±0.0057	0.8932±0.0023	0.8943±0.0083	0.9198±0.0015	0.9210±0.0052	0.9277±0.0071	0.9327±0.0014	**0.9356±0.0070**	**SSIM**

validation experiments are demonstrated in Figs. 5–7. The transition smoothing effect exhibited by SRCNN on the reconstruction task also appears in the robustness experiment results. Besides, VDSR and DSSR show obvious image degradation in the robustness validation experiments. CertSR and WGSR still maintain good edge reconstruction results. The proposed scheme achieves commendable evaluation in the visualization effect of robustness experiments.

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TABLE VI

ROBUSTNESS QUANTITATIVE EVALUATION OF THE PROPOSED MODEL AND DIFFERENT SCENARIOS USING THE UC MERCED DATASET AT SCALE = 4

Model / Scene Class	Bicubic	SRCNN	VDSR	PCRBPN	DSSR	CertSR	WGSR	Purposed	Scale ×4
Average of All	22.3026±0.6178	23.3331±0.4639	23.4848±0.7543	25.2078±0.4251	25.2032±0.5873	25.4299±0.5456	25.9420±0.4446	**26.2062±0.8503**	PSNR
	0.6540±0.0079	0.7385±0.0033	0.7350±0.0097	0.7626±0.0057	0.7637±0.0041	0.7729±0.0020	0.7755±0.0013	**0.7768±0.0030**	SSIM
agricultural	22.4407±0.6991	23.4967±0.3473	23.8628±0.5667	25.7218±0.2462	25.5030±0.8728	25.8946±0.9696	25.4939±0.5941	**26.6052±0.8828**	PSNR
	0.6567±0.0041	0.7381±0.0091	0.7378±0.0081	0.7690±0.0010	0.7670±0.0099	0.7645±0.0071	0.7695±0.0083	**0.7756±0.0082**	SSIM
baseballdiamond	22.4851±0.7497	23.2418±0.9333	23.7356±0.1644	25.2122±0.2337	25.9073±0.1293	25.5509±0.2898	25.9007±0.3116	**26.3056±0.4880**	PSNR
	0.5653±0.0035	0.7375±0.0027	0.7376±0.0048	0.7651±0.0028	0.7685±0.0031	0.7621±0.0050	0.7667±0.0088	**0.7702±0.0023**	SSIM
denseresidential	18.2805±0.2372	19.6070±0.4160	19.9177±0.3473	21.8205±0.8610	21.4268±0.5833	21.2572±0.7882	21.9374±0.9555	**22.4265±0.4290**	PSNR
	0.6014±0.0055	0.6443±0.0044	0.6460±0.0042	0.6968±0.0064	0.6939±0.0079	0.6995±0.0059	0.6966±0.0033	**0.7061±0.0073**	SSIM
golfcourse	25.9855±0.5881	27.4029±0.2709	27.7895±0.5778	29.6531±0.3858	29.3866±0.9953	29.4809±0.7385	29.9842±0.9828	**30.4687±0.5736**	PSNR
	0.7082±0.0058	0.8085±0.0015	0.8093±0.0096	0.8196±0.0033	0.8181±0.0077	0.8112±0.0084	0.8180±0.0096	**0.8215±0.0028**	SSIM
mediumresidential	19.1550±0.8399	20.2340±0.8614	20.3064±0.9332	21.8543±0.2358	21.1489±0.5140	21.3941±0.6866	21.5542±0.5860	**22.6930±0.7379**	PSNR
	0.32±0.0027	0.6769±0.0048	0.6732±0.0075	0.6995±0.0061	0.6981±0.0019	0.6941±0.0052	0.6984±0.0036	**0.7028±0.0079**	SSIM
overpass	24.3072±0.3688	25.2836±0.3587	25.6517±0.4338	26.5817±0.4709	26.1533±0.5218	26.4985±0.3215	**27.8442±0.8371**	27.2152±0.9070	PSNR
	0.6837±0.0030	0.7612±0.0012	0.7628±0.0063	0.7799±0.0043	0.7747±0.0014	0.7744±0.0029	0.7793±0.0075	**0.7825±0.0023**	SSIM
runway	24.9069±0.4887	25.9868±0.4987	25.7213±0.3640	27.6524±0.6713	27.9031±0.5091	27.5737±0.3587	27.7055±0.2056	**28.2917±0.7506**	PSNR
	0.6892±0.0029	0.7581±0.0070	0.7511±0.0095	0.7731±0.0044	0.7753±0.0040	0.7726±0.0027	0.7766±0.0061	**0.7887±0.0044**	SSIM
storagetanks	24.4975±0.4842	25.1721±0.7348	25.6586±0.9282	27.5651±0.8868	27.1842±0.5134	27.2560±0.4156	28.4356±0.2641	**28.4833±0.6668**	PSNR
	0.6819±0.0071	0.7560±0.0066	0.7540±0.0049	0.7787±0.0099	0.7764±0.0076	0.7721±0.0074	0.7793±0.0034	**0.7896±0.0075**	SSIM

TABLE VII

ROBUSTNESS QUANTITATIVE EVALUATION OF THE PROPOSED MODEL AND DIFFERENT SCENARIOS USING THE UC MERCED DATASET AT SCALE = 8

Model / Scene Class	Bicubic	SRCNN	VDSR	PCRBPN	DSSR	CertSR	WGSR	Purposed	Scale ×8
Average of All	18.3726±0.7018	20.2558±0.7502	20.8233±0.7517	21.7443±0.4272	21.9013±0.1561	21.6210±0.1118	22.5681±0.1817	**22.7035±0.3700**	PSNR
	0.4716±0.0021	0.5150±0.0017	0.5186±0.0059	0.5289±0.0077	0.5245±0.0070	0.5250±0.0019	0.5292±0.0079	**0.5355±0.0070**	SSIM
agricultural	18.3059±0.8385	20.9193±0.9728	20.2726±0.4565	21.6771±0.1962	21.4260±0.5453	21.5340±0.7941	21.6364±0.8579	**22.5460±0.4169**	PSNR
	0.4748±0.0041	0.5174±0.0047	0.5148±0.0034	0.5216±0.0084	0.5217±0.0037	0.5222±0.0064	0.2530±0.0065	**0.5345±0.0095**	SSIM
baseballdiamond	18.3474±0.7714	20.2726±0.9933	20.2449±0.3520	21.8571±0.1351	21.6447±0.7756	21.1907±0.2426	21.8170±0.3795	**22.8293±0.9991**	PSNR
	0.4721±0.0060	0.5173±0.0033	0.5155±0.0022	0.5255±0.0080	0.5252±0.0092	0.5264±0.0067	0.5255±0.0080	**0.5329±0.0062**	SSIM
denseresidential	15.2824±0.2561	17.4013±0.8320	17.2037±0.6900	17.2661±0.2547	17.5074±0.8790	17.5529±0.3255	17.3195±0.4543	**18.2968±0.2226**	PSNR
	0.4598±0.0086	0.4770±0.0022	0.4731±0.0036	0.4891±0.0080	0.4892±0.0039	0.4863±0.0030	0.4883±0.0017	**0.4956±0.0034**	SSIM
golfcourse	23.7188±0.7995	25.6530±0.7015	25.9701±0.5375	25.1454±0.6748	25.9467±0.1458	26.5664±0.5809	**26.7258±0.5414**	26.5627±0.7801	PSNR
	0.5061±0.0034	0.5720±0.0049	0.5785±0.0020	0.5797±0.0071	0.5723±0.0065	0.5843±0.0090	**0.5867±0.0080**	0.5856±0.0025	SSIM
mediumresidential	15.1996±0.3539	17.7120±0.4261	17.6890±0.9396	17.7046±0.7254	17.7448±0.1075	17.6997±0.4359	17.3607±0.2896	**18.1443±0.5979**	PSNR
	0.4551±0.0067	0.4923±0.0052	0.4965±0.0019	0.4958±0.0094	0.4998±0.0029	0.4825±0.0076	0.4843±0.0035	**0.4934±0.0051**	SSIM
overpass	19.7246±0.8768	21.9917±0.9670	22.5005±0.5956	22.6384±0.6912	22.8014±0.9092	22.8655±0.6791	22.8054±0.2967	**23.1226±0.6860**	PSNR
	0.4892±0.0096	0.5273±0.0063	0.5323±0.0062	0.5326±0.0043	0.5461±0.0019	**0.5462±0.0039**	0.5365±0.0083	0.5424±0.0072	SSIM
runway	20.3648±0.3163	22.8050±0.8377	22.2600±0.6229	23.8325±0.9468	23.2228±0.6150	23.9221±0.3150	23.8118±0.2012	**24.1545±0.6143**	PSNR
	0.4897±0.0054	0.5253±0.0033	0.5321±0.0080	0.5441±0.0058	0.5484±0.0071	0.5450±0.0086	0.5464±0.0042	**0.5597±0.0013**	SSIM
storagetanks	20.1524±0.3219	22.7354±0.2915	22.7928±0.8189	23.7295±0.7443	23.7256±0.2995	23.6843±0.7321	23.6319±0.3357	**24.3142±0.1107**	PSNR
	0.4831±0.0017	0.5333±0.0080	0.5311±0.0077	0.5419±0.0069	0.5472±0.0024	0.5514±0.0049	0.5518±0.0091	**0.5561±0.0068**	SSIM

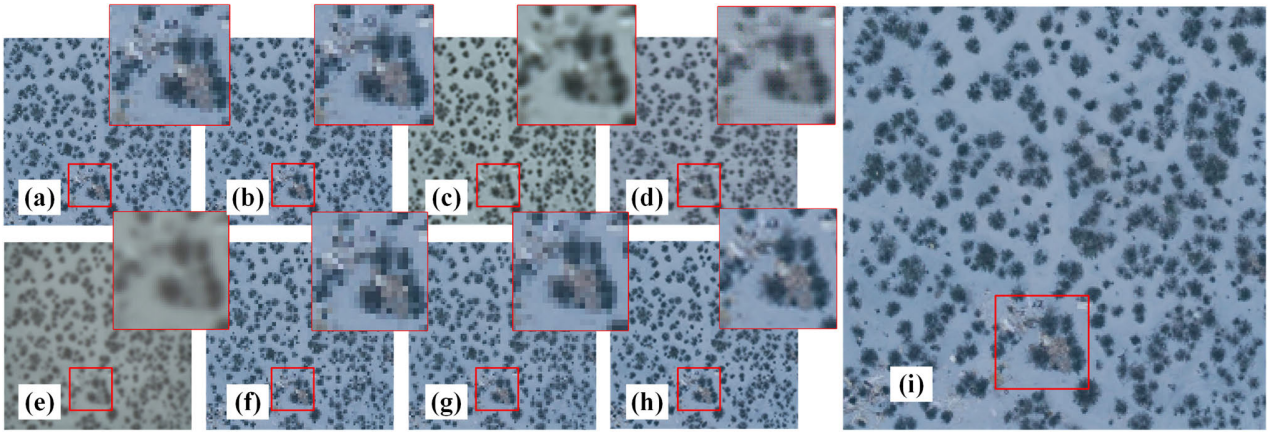


Fig. 5. Robustness experiments on the NWPU-RESISC45 dataset at Scale = 2. The figure is shown as chaparral. (a) Bicubic, (b) SRCNN, (c) VDSR, (d) DSSR, (e) PCRBPN, (f) CertSR, (g) WGSR, (h) proposed model, and (i) reference image.

D. Ablation Experiments

This section unfolds the ablation experiments and analyses for each design option in the proposed model based on two datasets.

1) Quantitative Analysis: Table VIII presents the performance evaluation metrics for each module ablation experiment of the proposed model (DDOM). While the complex deep network (CDN) exhibits limitations in recovering frequency-domain data, the combined approach of RSTN in the image domain and CDN in the frequency domain surpasses single-domain network designs in performance. Moreover, introducing cross-domain data consistency constraints on the image-domain RSTN significantly enhances model reconstruction performance.

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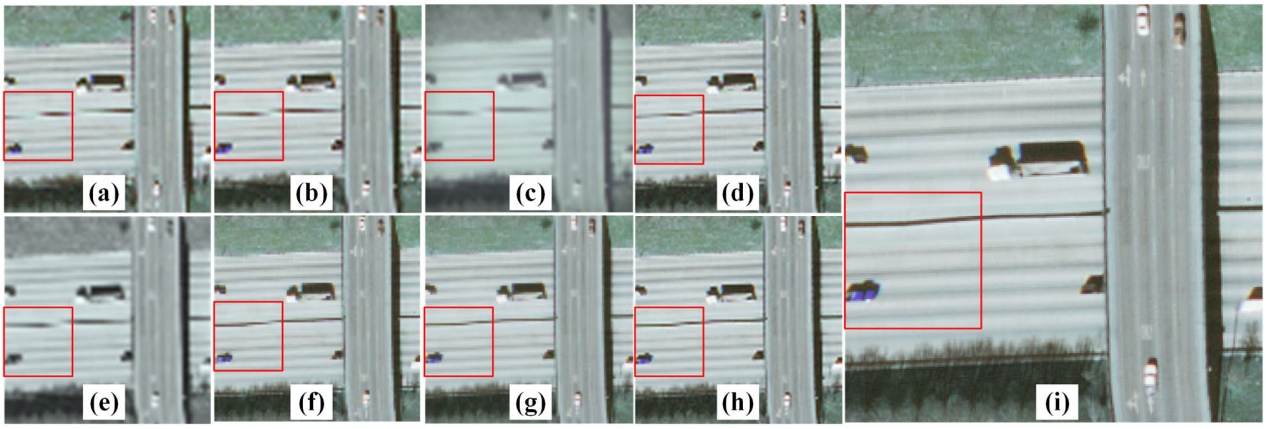


Fig. 6. Robustness experiments on the NWPU-RESISC45 dataset at Scale = 4. The figure is shown as overpass. (a) Bicubic, (b) SRCNN, (c) VDSR, (d) DSSR, (e) PCRBPN, (f) CertSR, (g) WGSR, (h) proposed model, and (i) reference image.

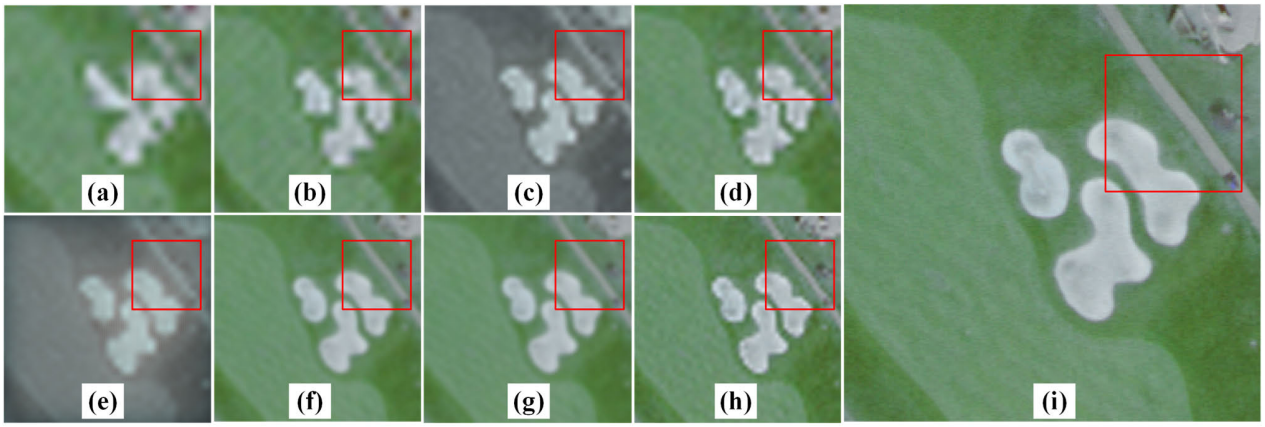


Fig. 7. Robustness experiments on the NWPU-RESISC45 dataset at Scale = 8. The figure is shown as golfcourse. (a) Bicubic, (b) SRCNN, (c) VDSR, (d) DSSR, (e) PCRBPN, (f) CertSR, (g) WGSR, (h) proposed model, and (i) reference image.

TABLE VIII

QUANTITATIVE RESULTS (MEAN ± STANDARD DEVIATION) OF ABLATION EXPERIMENTS ASSESSMENT BASED ON THE NWPU-RESISC45 DATASET, WITH BOLD MARKINGS INDICATING THE BEST VALUES

Module				Scale × 2		Scale × 4		Scale × 8	
CDN	Operator	RSTN	DCC	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
✓	✓			29.4324±0.3411	0.8861±0.0102	22.9737±0.5240	0.7185±0.0053	18.8196±0.7769	0.4595±0.0120
		✓		30.4131±0.4613	0.8921±0.0064	23.7362±0.7082	0.7228±0.0076	19.1107±0.6983	0.4879±0.0065
		✓	✓	32.2944±0.8395	0.9253±0.0063	25.1437±0.7433	0.7589±0.0041	21.8979±0.6399	0.5168±0.0085
✓	✓	✓		31.5162±0.4613	0.9095±0.0113	24.6961±0.4099	0.7480±0.0073	20.7014±0.1205	0.5080±0.0058
✓	✓	✓	✓	**33.5356±0.1322**	**0.9338±0.0101**	**26.2062±0.8503**	**0.7768±0.0030**	**22.7035±0.3700**	**0.5355±0.0070**

2) Data Consistency Constraints: Experimental validation was conducted to explore the relationship between cross-domain data consistency constraints and the performance of the proposed model (DDOM). The results are visualized in Fig. 8, demonstrating that the proposed model with data consistency constraint exhibits minimal pixel data distortion and superior fidelity in detail reconstruction compared to the control group. Fig. 9 illustrates the loss between the reconstructed results and the reference image, with the control group showing more pronounced local and overall loss. This underscores how the data consistency constraint enhances the data fidelity of the proposed model in reconstruction tasks.

3) Number of RSTBs: The number of RSTBs is an important attribute that affects the structure and volume of RSTN. In this section, experiments are designed around exploring the relationship between the number of RSTBs and the performance of the proposed model (DDOM). The performance evaluation and robustness of the proposed model in various reconstruction task scenarios with incremental RSTB are quantitatively evaluated in Figs. 10 and 11. Since the number of RSTBs is incremented to 4, it is not obvious that continuous incremental increase in the number of RSTBs can improve the performance of the proposed model. The tradeoff between resource loss and performance of the proposed model drives us to set the number of RSTB to 4. When four RSTBs are

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TABLE IX
RESOURCE LOSS AND MEMORY FOOTPRINT OF THE PROPOSED MODEL AND SCENARIOS

Indicators	Bicubic	SRCNN	VDSR	PCRCAN	DSSR	CertSR	WGSR	Purposed
FLOPs(G, Scale=2)	-	20.3	-	90.8	107.5	-	103.11	121.03
Param(M)	-	0.7	5.5	3.28	60	59.8	57	61.78
TIMES(ms)	-	15	65	61	243	107	252	280

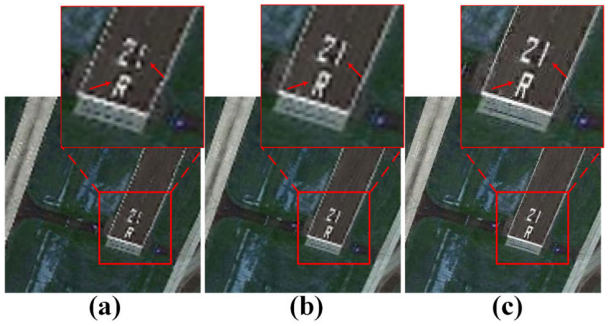


Fig. 8. Plot of the effect of consistency constraint module validation experiments performed under the NWPU-RESISC45 dataset at Scale = 2. The figure shows the runway. (a) Proposed model without consistency constraints (control group). (b) Proposed model with consistency constraints. (c) Reference image.

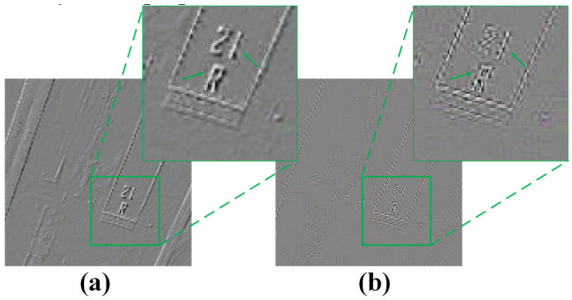


Fig. 9. Validation loss effect plot of the consistency constraint module performed under the NWPU-RESISC45 dataset. (a) Proposed model without consistency constraints (control group). (b) Proposed model with consistency constraints.

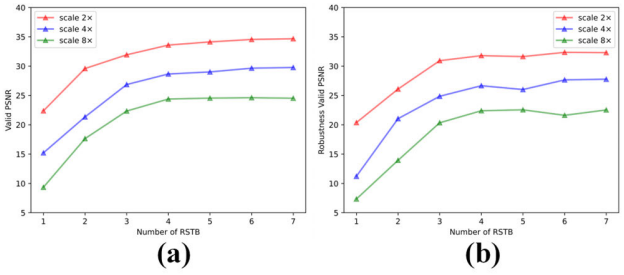


Fig. 10. Validation of the effect of the number of RSTB on the PSNR of the model reconstructed images. (a) Experimental results based on NWPU-RESISC45 dataset. (b) Experimental results based on UC Merced dataset.

included in the RSTN, the model shows a significant improvement in reconstruction performance and is more acceptable in terms of memory resource loss.

E. Downstream Task

In this section, we evaluate the effectiveness of the SR processed images for downstream tasks. Specifically, our

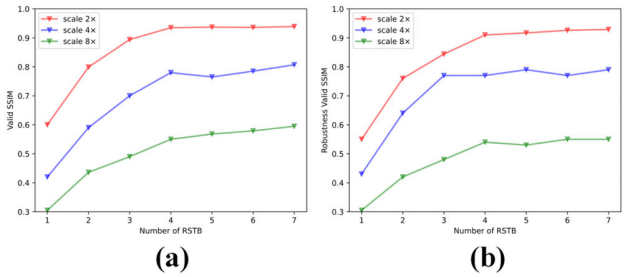


Fig. 11. Validation of the effect of the number of RSTB on the SSIM of the model reconstructed images. (a) Experimental results based on NWPU-RESISC45 dataset. (b) Experimental results based on UC Merced dataset.

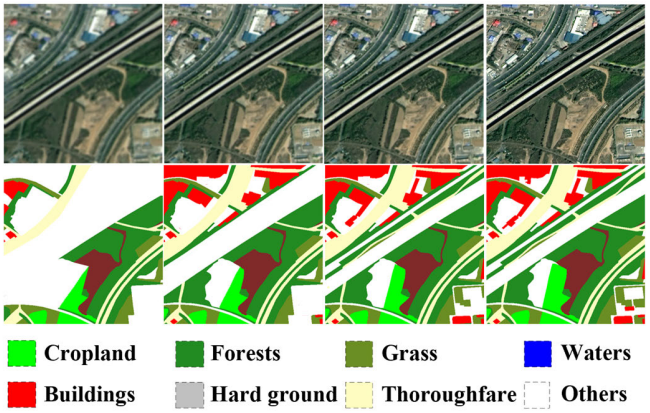


Fig. 12. (From left to right) LR image, the image processed by CertSR and the proposed model, the reference image, and the result after doing semantic segmentation processing on these images in turn.

experiments focus on 2-m resolution remote sensing images of the Taiyuan area in China, which includes both urban and suburban regions. We compare the results of the proposed model with those of other models in the context of semantic segmentation of remote sensing images. In terms of experimental strategy, we initially processed the LR image data using the proposed model and CertSR. Subsequently, the outcomes were employed in a remote sensing image parcel extraction platform that incorporates a semantic segmentation module, with the results depicted in Fig. 12. The segmentation tasks include terrain extraction and image classification. The results show that the proposed model has significant potential to enhance the performance of downstream tasks.

IV. DISCUSSION

The computational resource loss of the model is of concern. As a common drawback in deep learning models, in this section, the computational resource loss of the proposed model (DDOM) versus other schemes is evaluated and briefly analyzed. The quantitative results of the evaluation are shown in

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Table IX. The data suggest that the computational performance of the proposed model is more acceptable compared to the other models with the existing reconstruction performance of the proposed model.

V. Conclusion and Outlook

In this article, we design the frequency-domain transformation operator for fused DFT. Based on the operator, a DDOM is designed for SR reconstruction of remote sensing images. In the frequency-domain task, DDOM uses a deep complex network, which performs well in the spectral and phase information recovery work. In the image-domain task, ddom uses a Swin Transformer network with residuals, which shows exceptional performance in image resolution enhancement work. For the image-domain model, the dual-domain data consistency constraint not only improves the distortion of image detail reconstruction by introducing the frequency-domain loss but also guides the model to maintain positivity during the training optimization process from the cross-domain dimension. Extensive experiments show that both qualitative and quantitative results of the proposed model on the LR remote sensing image reconstruction task are due to the existing technical solutions. Moreover, the reconstruction results on datasets not involved in training confirm the robustness of the proposed model. This article provides new ideas and solutions to the problem of insufficient resolution of images due to enlarged size. Meanwhile, the idea of data consistency constraints will have the potential to promote the development of other remote sensing image research areas (e.g., anomaly detection and target extraction). The establishment of cross-domain target optimization is an important guarantee for the working effect of the proposed model, but there is still a need for further research on the design of the frequency-domain subnetwork of the proposed model. This study only validates a single cross-domain approach based on the DFT, and we cannot ascertain whether the principles of domain transformation and domain alignment exhibit generalizability in the context of remote sensing image processing. In the future, we will continue to seek the possibility of pixel spatial objective optimization on the task of remote sensing image frequency-domain data recovery around the problem of consistency preservation across data domains.

References

[1] M. F. Goodchild and Y. Shiren, "A hierarchical spatial data structure for global geographic information systems," *CVGIP: Graph. Models Image Process.*, vol. 54, no. 1, pp. 31–44, Jan. 1992, doi: 10.1016/1049-9652(92)90032-s.

[2] C. Toth and G. Jóźków, "Remote sensing platforms and sensors: A survey," *ISPRS J. Photogramm. Remote Sens.*, vol. 115, pp. 22–36, May 2016, doi: 10.1016/j.isprsjprs.2015.10.004.

[3] R. Olivier and C. Hanqiang, "Nearest neighbor value interpolation," *Int. J. Adv. Comput. Sci. Appl.*, vol. 3, no. 4, pp. 25–30, Apr. 2012, doi: 10.14569/IJACSA.2012.030405.

[4] M. Mastyło, "Bilinear interpolation theorems and applications," *J. Funct. Anal.*, vol. 265, no. 2, pp. 185–207, Jul. 2013, doi: 10.1016/j.jfa.2013.05.001.

[5] Z. Dengwen, "An edge-directed bicubic interpolation algorithm," in *Proc. 3rd Int. Congr. Image Signal Process.*, Yantai, China, Oct. 2010, pp. 1186–1189, doi: 10.1109/CISP.2010.5647190.

[6] J. Liu, Z. Gan, and X. Zhu, "Directional bicubic interpolation—A new method of image super-resolution," presented at the 3rd Int. Conf. Multimedia Technol. (ICMT), Guangzhou, China, 2013, doi: 10.2991/icmt-13.2013.57.

[7] P. Xia et al., "Performance comparison of bilinear interpolation, bicubic interpolation, and B-spline interpolation in parallel phase-shifting digital holography," *Opt. Rev.*, vol. 20, no. 2, pp. 193–197, Mar. 2013, doi: 10.1007/s10043-013-0033-2.

[8] A. Amanatiadis and I. Andreadis, "A survey on evaluation methods for image interpolation," *Meas. Sci. Technol.*, vol. 20, no. 10, Oct. 2009, Art. no. 104015, doi: 10.1088/0957-0233/20/10/104015.

[9] N. Bleistein, R. A. Handelsman, and J. S. Lew, "Functions whose Fourier transforms decay at infinity: An extension of the riemann-lebesgue lemma," *SIAM J. Math. Anal.*, vol. 3, no. 3, pp. 485–495, Aug. 1972, doi: 10.1137/0503046.

[10] H. Tao, X. Tang, J. Liu, and J. Tian, "Superresolution remote sensing image processing algorithm based on wavelet transform and interpolation," in *Proc. Image Process. Pattern Recognit. Remote Sens.*, Hangzhou, China, Jun. 2003, pp. 259–263, doi: 10.1117/12.468078.

[11] G. Hong and Y. Zhang, "Wavelet-based image registration technique for high-resolution remote sensing images," *Comput. Geosci.*, vol. 34, no. 12, pp. 1708–1720, Dec. 2008, doi: 10.1016/j.cageo.2008.03.005.

[12] W. Jin-liang, H. Zeng, W. Yan-ying, and G. Liu, "Super resolution reconstruction of TM remote sensing image based on wavelet analysis," *Remote Sens. Technol. Appl.*, vol. 31, no. 3, pp. 476–480, Jul. 2016.

[13] J. P. Papa, L. M. G. Fonseca, and L. A. S. de Carvalho, "Projections onto convex sets through particle swarm optimization and its application for remote sensing image restoration," *Pattern Recognit. Lett.*, vol. 31, no. 13, pp. 1876–1886, Oct. 2010, doi: 10.1016/j.patrec.2010.02.012.

[14] G. Ferraiuolo, V. Pascazio, and G. Schirinzi, "Maximum *a posteriori* estimation of height profiles in InSAR imaging," *IEEE Geosci. Remote Sens. Lett.*, vol. 1, no. 2, pp. 66–70, Apr. 2004, doi: 10.1109/LGRS.2003.822882.

[15] Y. Zhang, A. Duijster, and P. Scheunders, "A Bayesian restoration approach for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 9, pp. 3453–3462, Sep. 2012, doi: 10.1109/TGRS.2012.2184122.

[16] P. Rasti, H. Demirel, and G. Anbarjafari, "Iterative back projection based image resolution enhancement," in *Proc. 8th Iranian Conf. Mach. Vis. Image Process. (MVIP)*, Zanjan, Iran, Sep. 2013, pp. 237–240, doi: 10.1109/IRANIANMVIP.2013.6779986.

[17] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2016, doi: 10.1109/TPAMI.2015.2439281.

[18] L. Liebel and M. Körner, "Single-image super resolution for multispectral remote sensing data using convolutional neural networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 41, pp. 883–890, Jun. 2016, doi: 10.5194/isprsarchives-XLI-B3-883-2016.

[19] J. Kim, J. K. Lee, and K. M. Lee, "Accurate image super-resolution using very deep convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 1646–1654, doi: 10.1109/CVPR.2016.182.

[20] X. Qin, X. Gao, and K. Yue, "Remote sensing image super-resolution using multi-scale convolutional neural network," in *Proc. 11th U.K.-Europe-China Workshop Millim. Waves Terahertz Technol. (UCMMT)*, HangZhou, China, Sep. 2018, pp. 1–3, doi: 10.1109/UCMMT45316.2018.9015801.

[21] W. Xu, G. Xu, Y. Wang, X. Sun, D. Lin, and Y. Wu, "Deep memory connected neural network for optical remote sensing image restoration," *Remote Sens.*, vol. 10, no. 12, p. 1893, Nov. 2018, doi: 10.3390/rs10121893.

[22] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778, doi: 10.1109/CVPR.2016.90.

[23] Z. Pan, W. Ma, J. Guo, and B. Lei, "Super-resolution of single remote sensing image based on residual dense backprojection networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7918–7933, Oct. 2019, doi: 10.1109/TGRS.2019.2917427.

[24] W. Ma, Z. Pan, J. Guo, and B. Lei, "Achieving super-resolution remote sensing images via the wavelet transform combined with the recursive res-net," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3512–3527, Jun. 2019, doi: 10.1109/TGRS.2018.2885506.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:59:18 UTC from IEEE Xplore. Restrictions apply.

[25] J. M. Haut, R. Fernandez-Beltran, M. E. Paoletti, J. Plaza, and A. Plaza, "Remote sensing image superresolution using deep residual channel attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9277–9289, Nov. 2019, doi: 10.1109/TGRS.2019.2924818.

[26] D. Zhang, J. Shao, X. Li, and H. T. Shen, "Remote sensing image super-resolution via mixed high-order attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5183–5196, Jun. 2021, doi: 10.1109/TGRS.2020.3009918.

[27] S. Zhang, Q. Yuan, J. Li, J. Sun, and X. Zhang, "Scene-adaptive remote sensing image super-resolution using a multiscale attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4764–4779, Jul. 2020, doi: 10.1109/TGRS.2020.2966805.

[28] Y. Li, Y. Iwamoto, L. Lin, and Y.-W. Chen, "Parallel-connected residual channel attention network for remote sensing image super-resolution," in *Proc. Asian Conf. Comput. Vis. (ACCV Workshops)* (Lecture Notes in Computer Science), vol. 12628, I. Sato and B. Han, Eds., Cham, Switzerland: Springer, 2021, pp. 18–30, doi: 10.1007/978-3-030-69756-3_2.

[29] C. Ledig et al., "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 105–114, doi: 10.1109/CVPR.2017.19.

[30] M. Lavreniuk, L. Shumilo, and A. Lavreniuk, "Generative adversarial networks for the satellite data super resolution based on the transformers with attention," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Pasadena, CA, USA, Jul. 2023, pp. 6294–6297, doi: 10.1109/igarss52108.2023.10281826.

[31] A. Mohammed, M. Kashif, M. H. Zama, M. A. Ansari, and S. Ali, "Master GAN: Multiple attention is all you need: A multiple attention guided super resolution network for dems," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Pasadena, CA, USA, Jul. 2023, pp. 5154–5157, doi: 10.1109/igarss52108.2023.10283196.

[32] Y. Wang, C. M. Albrecht, N. A. A. Braham, L. Mou, and X. X. Zhu, "Self-supervised learning in remote sensing: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 4, pp. 213–247, Dec. 2022, doi: 10.1109/MGRS.2022.3198244.

[33] H. Liu, R. V. Vinu, Z. Chen, J. Liang, and J. Pu, "Compact snapshot phase-shifting digital holographic imaging systems using pixelated polarization camera," in *Coded Optical Imaging*, J. Liang, Ed., Cham, Switzerland: Springer, 2024, pp. 683–704, doi: 10.1007/978-3-031-39062-3_37.

[34] Y. Yang and S. Soatto, "FDA: Fourier domain adaptation for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 4084–4094, doi: 10.1109/CVPR42600.2020.00414.

[35] Q. Xu, R. Zhang, Y. Zhang, Y. Wang, and Q. Tian, "A Fourier-based framework for domain generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Nashville, TN, USA, Jun. 2021, pp. 14378–14387, doi: 10.1109/CVPR46437.2021.01415.

[36] C. Trabelsi et al., "Deep complex networks," 2017, *arXiv:1705.09792*.

[37] N. Guberman, "On complex valued convolutional neural networks," 2016, *arXiv:1602.09046*.

[38] M. Cogswell, F. Ahmed, R. Girshick, L. Zitnick, and D. Batra, "Reducing overfitting in deep networks by decorrelating representations," 2015, *arXiv:1511.06068*.

[39] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[40] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van G., and R. Timofte, "SwinIR: Image restoration using Swin transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV) Workshops*, Montreal, BC, Canada, Oct. 2021, pp. 1833–1844, doi: 10.1109/ICCVW54120.2021.00210.

[41] Y. Cheng et al., "LIR-net: Learnable iterative reconstruction network for fan beam CT sparse-view reconstruction," *IEEE Trans. Comput. Imag.*, vol. 10, pp. 181–195, 2024, doi: 10.1109/TCI.2024.3358673.

[42] L. Greengard, J.-Y. Lee, and S. Inati, "The fast sinc transform and image reconstruction from nonuniform samples ink-space," *Commun. Appl. Math. Comput. Sci.*, vol. 1, no. 1, pp. 121–131, Dec. 2006, doi: 10.2140/camcos.2006.1.121.

[43] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst.*, San Jose, CA, USA, Nov. 2010, pp. 270–279, doi: 10.1145/1869790.1869829.

[44] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017, doi: 10.1109/JPROC.2017.2675998.

[45] F. Russo, "A method for estimation and filtering of Gaussian noise in images," *IEEE Trans. Instrum. Meas.*, vol. 52, no. 4, pp. 1148–1154, Aug. 2003, doi: 10.1109/TIM.2003.815989.

[46] X. Dong, X. Sun, X. Jia, Z. Xi, L. Gao, and B. Zhang, "Remote sensing image super-resolution using novel dense-sampling networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1618–1633, Feb. 2021, doi: 10.1109/TGRS.2020.2994253.

[47] J. Zhang et al., "Pan-sharpening with wavelet-enhanced high-frequency information," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5402914, doi: 10.1109/TGRS.2024.3367165.

[48] Z. Chaouai and M. Tamaazousti, "Universal robustness via median randomized smoothing for real-world super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2024, pp. 9059–9068, doi: 10.1109/cvpr52733.2024.00865.

[49] C. Korkmaz, A. M. Tekalp, and Z. Dogan, "Training generative image super-resolution models by wavelet-domain losses enables better control of artifacts," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2024, pp. 5926–5936, doi: 10.1109/cvpr52733.2024.00566.

[50] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multi-stage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117, doi: 10.1109/TGRS.2024.3391014.

[51] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509417, doi: 10.1109/TGRS.2023.3267890.

[52] J. Li, K. Zheng, W. Liu, Z. Li, H. Yu, and L. Ni, "Model-guided coarse-to-fine fusion network for unsupervised hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023, doi: 10.1109/LGRS.2023.3309854.

[53] J. Li, K. Zheng, J. Yao, L. Gao, and D. Hong, "Deep unsupervised blind hyperspectral and multispectral data fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2022.3151779.

[54] H. Yu, Z. Ling, K. Zheng, L. Gao, J. Li, and J. Chanussot, "Unsupervised hyperspectral and multispectral image fusion with deep spectral–spatial collaborative constraint," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534114, doi: 10.1109/TGRS.2024.3472226.

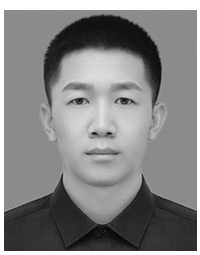
[55] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317, doi: 10.1109/TGRS.2023.3300043.

[56] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218, doi: 10.1109/TGRS.2025.3531646.



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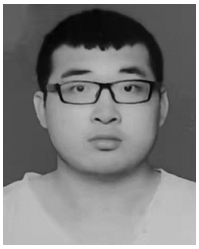
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